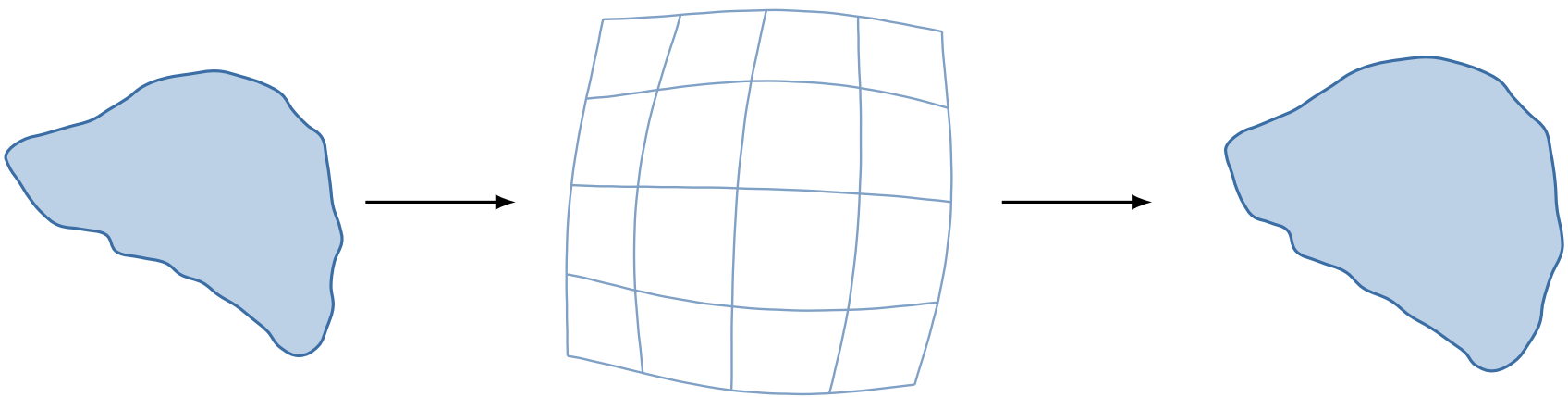S_0 φ $\varphi(S_0)$